Haotian WEI

Ph.D. Candidate, 7+ yrs Theoretical Quantum Scientist
weihaotian776@gmail.com ◦ Houston, TX ◦ github.com/htwei17 (800+★) ◦ www.linkedin.com/in/htwei17

EDUCATION

Ph. D. in Physics, Rice University, *Houston, TX* — Exp. Aug 2026
M. Sc. in Physics, Rice University, *Houston, TX* — Dec 2024
B.Sc. in Physics, Fudan University, *Shanghai, China* — Jun 2020

- Outstanding Graduate of Class 2020

Visiting Student, University of California, *Berkeley, CA* — Aug - Dec 2017

PUBLICATIONS

Highlights: 7 papers **| 320+** citations | **70+** media coverages | **3** Editors' Suggestions | Nat. Phys,, PRL, PRXQ, Sci. Adv.
Google Scholar: scholar.google.com/citations?user=mu--7-UAAAAJ

- Zheyuan Li, Rupsa De, Rishi Sivakumar, William Huie, **Hao-Tian Wei**, Justin D. Piel, Chris H. Greene, Kaden R. A. Hazzard, Zoe Z. Yan, Jacob P. Covey. "Quantum science with arrays of metastable helium-3 atoms" arXiv:2601.06763 (2026). *[PRX Quantum Accepted]*
- Dasom Kim, Sohail Dasgupta, Xiaoxuan Ma, Joong-Mok Park, **Hao-Tian Wei**, Liang Luo, Jacques Doumani, Xinwei Li, Wanting Yang, Di Cheng, Richard H. J. Kim, Henry O. Everitt, Shojiro Kimura, Hiroyuki Nojiri, Jigang Wang, Shixun Cao, Motoaki Bamba, Kaden R. A. Hazzard, and Junichiro Kono. "Observation of the magnonic Dicke superradiant phase transition" *Science Advances* **11**, eadt1691 (2025). *[Featured in news]*
- **Hao-Tian Wei**, Eduardo Ibarra-García-Padilla, Michael L. Wall, and Kaden R. A. Hazzard. "Hubbard Parameters for Programmable Tweezer Arrays" *Physical Review A* **109**, 013318 (2024). *[Editors' Suggestion]*
- Zoe Z. Yan, Benjamin M. Spar, Max L. Prichard, Sungjae Chi, **Hao-Tian Wei**, Eduardo Ibarra-García-Padilla, Kaden R. A. Hazzard, and Waseem S. Bakr. "Two-Dimensional Programmable Tweezer Arrays of Fermions" *Physical Review Letters* **129**.123201 (2022). *[Editors' Suggestion] [Featured in news]*
- Shintaro Taie, Eduardo Ibarra-García-Padilla, Naoki Nishizawa, Yosuke Takasu, Yoshihito Kuno, **Hao-Tian Wei**, Richard T. Scalettar, Kaden R. A. Hazzard, and Yoshiro Takahashi. "Observation of Antiferromagnetic Correlations in an Ultracold SU(N) Hubbard Model" *Nature Physics* **18**.1356–61 (2022). *[Featured in news]*
- Eduardo Ibarra-García-Padilla, Sohail Dasgupta, **Hao-Tian Wei**, Shintaro Taie, Yoshiro Takahashi, Richard T. Scalettar, and Kaden R. A. Hazzard. "Universal Thermodynamics of an SU(N) Fermi-Hubbard Model" *Physical Review A* **104**.043316 (2021). *[Editors' Suggestion]*
- Yuan D. Liao, Han Li, Zheng Yan, **Hao-Tian Wei**, Wei Li, Yang Qi, and Zi Y. Meng. "Phase Diagram of the Quantum Ising Model on a Triangular Lattice under External Field" *Physical Review B* **103**.104416 (2021).

CONFERENCE PRESENTATIONS & TALKS

Invited talks:

- "A universal and efficient hybrid analog-digital fermionic quantum simulator", *IFF-CSIC* 2026*, Virginia Tech 2026*.

Oral & poster presentations:

- "A universal and efficient fermionic variational quantum simulator", *APS Global Physics Summit* 2026, *ITAMP Winter School* 2025, *eQMA Spring School & Fall Workshop* 2025, *QuantIPS 2025*, *APS DAMOP Meeting* 2025, *Texas Quantum Summit 2025*.
- "Fermionic programmable quantum simulators running variational algorithms", *APS DAMOP Meeting* 2024.
- "Parameters and algorithms for programmable Fermi-Hubbard tweezer arrays for quantum simulations", *Rice Quantum Group Meeting* 2023.
- "Effective Hubbard parameters for programmable tweezer arrays", *APS March Meeting* 2023.
- "Programmable Hubbard model in tweezer arrays", *RCQM Workshop* 2022, *QuantIPS* 2023.
- "Hubbard parameters of optical tweezer arrays in arbitrary 1- and 2-D geometries", *IUPAP Conference on Computational Physics* 2022.
- "Stroboscopic fermion tweezer arrays: heating and Hubbard parameters", *APS DAMOP Meeting* 2022.

PROFESSIONAL SKILLS

Physics Techniques: Quantum Dynamics, Quantum Criticality, Fermi-Hubbard, RG, Variational Methods, Lie Algebra Representation Theory, Lieb-Robinson Bound, Mean Field Theory
Quantum Hardware: Neutral Atom, Ion Traps, Superconducting Qubits
Programming Languages: 7+ yr Python (NumPy/SciPy/PyTorch/Qiskit), 4+ yr C/C++, MATLAB, Mathematica & Julia
Algorithms: Variational Quantum Algorithms, Optimization, Tensor Networks, Matrix Diagonalization (full /Arnoldi/Lanczos), Monte Carlo, Numerical Analysis, Data Correlation Analysis, Regression, Deep Neural Networks

ML/DL: Coursera Deep Learning Specialization, Reinforcement Learning Course, neural-network design & training, LLM prompt design & engineering, regressions, SGD/Adam/BFGS optimizers, statistics (SSR/SST/Pearson R), SQL
Application Skills: High-performance computing (HPC), Git workflow, data visualization (Matplotlib/Inkscape), MySQL
Open-Source Contributions: GitHub profile github.com/htwei17

- *Pymanopt (***800+★**) – extended core functionality of Riemannian manifold optimizer
- *HubbardTweezer* – individually developed gate-engineering toolkit of next-gen neutral-atom quantum hardware
- *Rice-thesis-latex* – restructured and modernized the most popular Rice University thesis LaTeX template

Languages: English (fluent), Chinese Mandarin (native)
Interests: Linguistics (phonetics/phonology), Saxophone

LEADERSHIP & OUTREACH

Physics & Astronomy Graduate Student Association (PAGSA), Rice University, *Houston, TX* Jun 2022 – Jun 2024

- Treasurer & Journal club coordinator

Academic Reviewer, Physical Review X, Physical Review Letters, Physical Review Research, Physical Review Applied, Physical Review A&B, European Physics Journal Plus Aug 2024 - Present

- Reviewed **20+** top research articles

APS March Meeting 2023, *Las Vegas, NV* Mar 2023

- Chaired a session in the world's largest physics conference with 14,000+ attendees